

Trainings ▾

Leak Test

⚠ WARNING ⚠

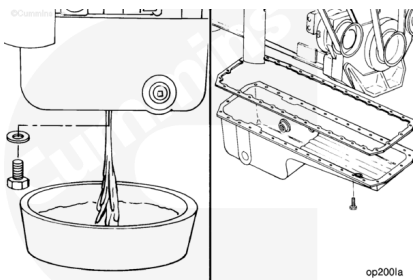
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

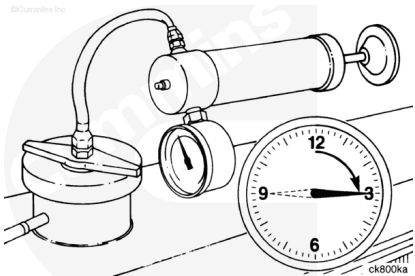
Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html)

Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html)



Pressurize the engine cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)

Apply the air pressure 15 minutes before inspecting the cylinder liner seats. Be sure the system is holding air pressure before beginning inspection.



Measurements

	kpa	psi
Air Pressure:	138	20

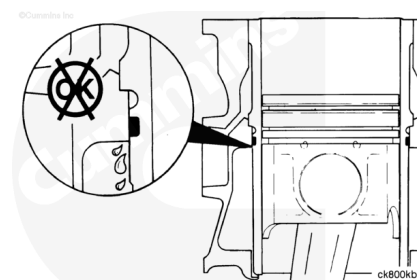
Inspect the outside diameter of the cylinder liners and the area below the cylinder liner seats in the cylinder block for coolant leaks.

If a leak is found:

- Remove the cylinder liner(s). Inspect the sealing ring and cylinder liner. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-028-tr.html>)

If a leak is found in the cylinder liner seat:

- Remove the cylinder liners. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-028-tr.html>) Inspect the cylinder block liner counterbore area. If fretting or wear is found, the cylinder block liner counterbore can be repaired. Refer to the Alternate Repair Manual, L10 Series Engines, Bulletin 3810310.



Install a new gasket and the lubricating oil pan. Refer to Procedure 007-025 in Section 7.

(</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)

Tighten the oil pan drain plug.

Torque Value: 88 n•m [65 ft-lb]

Fill the engine with clean 15W-40 oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)

Operate engine until it reaches a temperature of 80°C [180°F] and check for coolant or lubricating oil leaks.

If the malfunction resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14.

(</qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html>)

